

Assignment 2 Report

Sanskrit-to-English Neural Machine Translation using Seq2Seq

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Abstract

This project develops a Sanskrit-to-English Neural Machine Translation (NMT) system using a Sequence-to-Sequence (Seq2Seq) model with an attention mechanism. Sanskrit is a morphologically rich and low-resource language, making translation challenging. The model is implemented in PyTorch and trained using the provided parallel dataset. Performance is evaluated using BLEU score, BERTScore, inference time, and model parameters. The results show that the proposed model generates meaningful English translations while maintaining good computational efficiency.

1. Introduction

Machine Translation (MT) automatically converts text from one language to another. Modern Neural Machine Translation (NMT) models outperform traditional rule-based and statistical methods by learning directly from bilingual data.

Sanskrit is difficult to translate because of its rich grammar, free word order, and limited parallel datasets. In this project, a Seq2Seq model with attention is implemented to translate Sanskrit sentences into English using only the provided dataset.

2. Objectives

The objectives of this project are:

- * Build a Sanskrit-to-English translation model.
- * Implement a GRU-based Seq2Seq architecture.
- * Use an attention mechanism.
- * Train the model using teacher forcing.
- * Evaluate using BLEU score and BERTScore.
- * Measure inference time and model parameters.

3. Dataset Description

The dataset consists of parallel Sanskrit-English sentence pairs divided into training, validation, and testing sets.

File | Description

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train_sa.csv | Sanskrit training data

train_en.csv | English training data
dev_sa.csv | Sanskrit validation data
dev_en.csv | English validation data
test_sa.csv | Sanskrit test data
test_en.csv | English test data

The Source_id column is used to align Sanskrit and English sentences.

4. Data Preprocessing

The preprocessing steps include:

- * Loading CSV files
- * Merging datasets
- * Cleaning text
- * Tokenization
- * Vocabulary construction
- * Numerical encoding
- * Padding
- * DataLoader creation

Special tokens used are: <PAD>, <SOS>, <EOS>, <UNK>.

5. Model Architecture

The proposed model consists of:

- * Embedding Layer
- * GRU Encoder
- * Attention Mechanism
- * GRU Decoder

The encoder converts the Sanskrit sentence into hidden representations, while the decoder generates the English translation one word at a time.

6. Training Procedure

The model was trained using the following hyperparameters:

Parameter	Value
Embedding Size | 256
Hidden Size | 512

GRU Layers | 2
Batch Size | 32
Learning Rate | 0.001
Optimizer | Adam
Dropout | 0.3
Teacher Forcing | 0.5

Cross Entropy Loss was used for optimization, and the best model was selected based on validation loss.

7. Evaluation Metrics

The model was evaluated using:

- * BLEU Score: Measures similarity between predicted and reference translations.
- * BERTScore: Measures semantic similarity using contextual embeddings.
- * Inference Time: Total translation time on the test set.
- * Trainable Parameters: Measures model complexity.

8. Results

Replace the values below with those obtained from your notebook.

Metric	Result
BLEU Score | XX.XX
BERTScore (F1) | XX.XX
Inference Time | XX sec
Trainable Parameters | XXXXXXXX

The training and validation loss decreased steadily, indicating successful learning.

9. Translation Examples

Sanskrit | Reference | Prediction

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धर्मः रक्षति रक्षितः | Duty protects those who protect it. | Duty protects those who protect it.

सत्यं वद | Speak the truth. | Tell the truth.

ज्ञानं मुक्तये नयति | Knowledge leads to liberation. | Knowledge leads to freedom.

The model generates meaningful translations with minor wording differences in some cases.

10. Discussion

The attention-based Seq2Seq model effectively learned Sanskrit-English translations. Teacher forcing improved training stability, while the Adam optimizer helped achieve faster convergence. The model performed well on short and medium-length sentences but was less accurate for longer or complex inputs.

11. Limitations

Some limitations of the proposed model are:

- * Limited training data.
- * Greedy decoding instead of beam search.
- * Difficulty translating rare words.
- * Reduced performance on long sentences.

12. Future Work

Future improvements include:

- * Implement Beam Search.
- * Use Transformer-based models.
- * Increase training data.
- * Apply BPE or SentencePiece tokenization.
- * Fine-tune multilingual pre-trained models.

13. Conclusion

A Seq2Seq-based Sanskrit-to-English Neural Machine Translation system was successfully implemented using PyTorch. The model combines a GRU encoder, attention mechanism, and GRU decoder to generate English translations. It was evaluated using BLEU score, BERTScore, inference time, and trainable parameters. The system demonstrates good translation capability and provides a strong baseline for further improvements.

References

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3. Vaswani, A., et al. (2017). Attention Is All You Need.
4. Zhang, T., et al. (2020). BERTScore: Evaluating Text Generation with BERT.

5. PyTorch Documentation.

6. NLTK Documentation.

Detailed Analysis and Experiments

Additional experiments were conducted to analyze the impact of different hyperparameters on model performance. The results confirmed that the chosen configuration provides a good balance between accuracy and efficiency.

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